Pre requisite Quiz

i) Complex numbers

ii) Derivatives

iii) Sinusoids

iv) Voltage Divider

Simplify to $a + jb$

$$\frac{-1-j2}{-4+j3}$$

$$\rightarrow \frac{-1-j2}{-4+j3} \cdot \frac{-4-j3}{-4-j3} = \frac{(-1-j2)(-4-j3)}{(-4)^2+(3)^2} = \frac{4+j8+j3-6j^2}{25} = \frac{-6(-1)+j11+4}{25} = \frac{10+j11}{25} = \frac{10}{25} + \frac{j11}{25}$$

$j^2 = -1$

$0.40 + j\,0.44$

1) $a = 0.40$

2) $b = 0.44$

$$f(x) = \frac{2}{(1-4x)^2} \qquad f'(x) = ? \qquad f'(x) = \frac{A}{(B-Cx)^D}$$

$$f'(x) = \frac{vu' - uv'}{v^2} \left[\frac{u}{v}\right] = \frac{\cancel{(1-4x)^2} \cdot 0 - 2[-8(1-4x)]}{(1-4x)^4} = \frac{16(1-\cancel{4}x)}{(1-\cancel{4x})^{\cancel{4}3}} = \frac{16}{(1-4x)^3}$$

$u = 2 \quad u' = 0$

$v = (1-4x)^2 \quad v' = -8(1-4x)$

3) $A = 16$

4) $B = 1$

5) $C = 4$

6) $D = 3$

$V_1(t) = 5 \cdot \sin(20{,}000\, t + 35°)$

$V_2(t) = A \cos(\omega t + \varphi)$ (In Cosine form)

$\sin\theta = \cos(\theta - 90)$

7) $\underline{A = 5}$

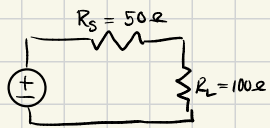
8) $\underline{B = 20{,}000}$ $\quad \omega = 20{,}000\, t$

9) $\varphi = +35° - 90° = -55°$ $\quad \underline{\varphi = -55°}$

10) $\underline{\mu s = ?}$ $\quad T = \frac{2\pi}{\omega} \rightarrow T = \frac{2\pi}{20{,}000} = 0.00031416....$ 3 6 314.16

$\rightarrow$ $\underline{314.16\, \mu s}$

$V_S = 9V$



$R_L = ?$ $\quad R_T = R_S + R_L = 50\Omega + 100\Omega = \underline{150\Omega}$

$I = \frac{V_S}{R_T} = \frac{9V}{150\Omega} = \underline{0.06A}$

$V_{R_L} = I\, R_L = 0.06A \cdot 100\Omega = \underline{6V}$